PHY202 Practice Exam 6 on chapter(s): 25, 26

1. $n_2 = 1.00 \ \because n_1 \sin\theta_1 = n_2 \sin\theta_2$
2. $\theta_c = 45.6^\circ \ \because n_1 \sin\theta_1 = n_2 \sin\theta_2$
3. $P = 43.5\,\mathrm{D} \ \because \frac{1}{d_0} + \frac{1}{d_i} = \frac{1}{f}, P = \frac{1}{f}$
4. $P = 50.8\,\mathrm{D} \ \because \frac{1}{d_0} + \frac{1}{d_i} = \frac{1}{f}, P = \frac{1}{f}$
5. $v = 1.88 \times 10^8 \frac{\mathrm{m}}{\mathrm{s}} \ \because n = \frac{c}{v}$
6. $P = 62.5\,\mathrm{D} \ \because P = \frac{1}{f}$
7. $m = -0.607 \ \because \frac{1}{d_0} + \frac{1}{d_i} = \frac{1}{f}, \frac{h_i}{h_0} = -\frac{d_i}{d_0} = m$
8. $\theta_2 = 6.45^\circ \ \because n_1 \sin\theta_1 = n_2 \sin\theta_2$
9. $h_i = -1.60 \times 10^{-5}\,\mathrm{m} \ \because \frac{h_i}{h_0} = -\frac{d_i}{d_0} = m$
10. $m = -3.00 \ \because \frac{1}{d_0} + \frac{1}{d_i} = \frac{1}{f}, \frac{h_i}{h_0} = -\frac{d_i}{d_0} = m$
11. $d_i = 0.150\,\mathrm{m} \ \because \frac{1}{d_0} + \frac{1}{d_i} = \frac{1}{f}$
12. $P = 1.72\,\mathrm{D} \ \because \frac{1}{d_0} + \frac{1}{d_i} = \frac{1}{f}, P = \frac{1}{f}$
13. $P = -4.81\,\mathrm{D} \ \because \frac{1}{d_0} + \frac{1}{d_i} = \frac{1}{f}, P = \frac{1}{f}$